

Eliza Hahn Neights

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Personal Statement

I am a high-energy astrophysicist studying the brief flashes of gamma rays produced by powerful jets that arise from events such as stellar death and neutron star mergers. I investigate the physical origins and processes that power these gamma-ray bursts (GRBs) using *Fermi*-GBM, for which I volunteer as a Burst Advocate. I explore the GRB science enabled by the upcoming COSI telescope and serve as Early Career Co-Lead for its GRB Science Team.

Education

- 2021-2027 **George Washington University** Ph.D in Physics (*in progress*)
Thesis: Studying Gamma-Ray Bursts with COSI and Fermi-GBM Analysis of a Record Ultra-long Gamma-Ray Burst
Advisors: Dr. Sylvain Guiriec & Dr. Carolyn Kierans
- 2021-2024 **George Washington University** M.Phil in Physics, M.S. in Physics
- 2017-2021 **Pennsylvania State University** B.S. in Astronomy & Astrophysics, B.S. in Physics
Minor in Mathematics
Advisors: Dr. Miguel Mostafá & Dr. Doug Cowen

Leadership Roles

- 2026-present **Early Career Co-Lead for COSI GRB Science Team**
First Early Career Co-Lead for a COSI science team and first student member of COSI Central Science Team. Manage >50 person group and organize monthly meetings.
- 2024-present **Burst Advocate for Fermi-GBM**
Volunteer for 6-8 shifts per month. Analyze triggered observations, ensure good data quality, and send alerts to scientific community.
- 2022-present **Spectral and Polarization Analysis Software Development for COSI**
Lead development of publicly available spectral and polarization analysis software used by scientific community to analyze COSI data. 🐙 github.com/cositools/cosipy
- 2023-2026 **Technical Lead for COSI GRB Science Team**
Led creation of simulated COSI transient observations used to develop analysis software, test onboard trigger algorithm, and make GRB science predictions. Developed software to simulate realistic population-level COSI observations of transients. Managed GRB team tasks, including development of COSI transient analysis pipelines. Planned and led monthly meetings.
🐙 github.com/cositools/grb-simulations

Awards & Scholarships

- 2026-2027 **Chateaubriand Fellowship**
Grant awarded by French Embassy's Office for Science and Technology to conduct GRB research using SVOM at Institute of Research in Astrophysics and Planetary Science in France for six months.
- 2025 **Gus W. Weiss Prize (Honorable Mention)**
Award celebrating graduate student research contributions and dedication to George Washington University Physics department and larger physics community.

Awards & Scholarships (continued)

- 2022-2025 **John Mather Nobel Scholarship**
Travel allowance awarded by John and Jane Mather Foundation for Science and the Arts towards cost of presenting research at conferences.
- 2017-2021 **Braddock Scholarship**
Four-year academic merit scholarship awarded by Penn State Eberly College of Science.
- 2019-2021 **M. Dean and Jean L. Underwood Scholarship in Physics**
Academic merit scholarship awarded by Penn State Eberly College of Science.
- 2019-2020 **Eberly College of Science Undergraduate Research Award**
Funding from Office of Science Engagement in Penn State Eberly College of Science to do undergraduate astrophysics research with Dr. Doug Cowen.
- 2017-2020 **Dean's List**
Maintained GPA above 3.5.
- 2018 **NASA Pennsylvania Space Grant**
Women in Science and Engineering Research scholarship to do undergraduate astrophysics research with Dr. Miguel Mostafá.

Publications

In addition to the publications below, I have led 14 GCN circulars and 1 Astronomer's Telegram.

E. Neights et al. 2026 was featured as a *Nature Astronomy* Research Highlight in Feb 2026.

Research profiles:  Orcid,  SciX,  Google Scholar

Refereed

E. Neights, E. Burns, C. L. Fryer, et. al., "GRB 250702B: discovery of a gamma-ray burst from a black hole falling into a star," *MNRAS*, Jan. 2026.  DOI: 10.1093/mnras/staf2019.

N. Parmiggiani, A. Bulgarelli, G. Panebianco, et. al. (E. Neights 5th out of 20 authors), "COSI Short Gamma-Ray Burst Localization Using BGO Shield Data," *ApJ*, Feb. 2026.  DOI: 10.3847/1538-4357/ae25fc.

A. Rizzo, N. Parmiggiani, A. Bulgarelli, et. al. (E. Neights 19th out of 21 authors), "Comparing Classical and Quantum Deep Learning Techniques for Anomaly Detection of Short-Duration Gamma-Ray Signals," *Astronomy and Computing*, Jul. 2026.  DOI: 10.1016/j.ascom.2026.101090.

A. C. Trigg, E. Burns, M. Negro, et. al. (E. Neights 14th out of 19 authors), "From rare events to a population: Discovering overlooked extragalactic magnetar giant flare candidates in archival Fermi Gamma-ray Burst Monitor data," *A&A*, Jun. 2026.  DOI: 10.1051/0004-6361/202558217.

H. Yoneda, T. Siebert, I. Martinez-Castellanos, et. al. (E. Neights 12th out of 26 authors), "Enhancing Compton telescope imaging with maximum a posteriori estimation: A modified Richardson–Lucy algorithm for the Compton Spectrometer and Imager," *A&A*, May 2025.  DOI: 10.1051/0004-6361/202453528.

H. C. Gulick, E. Neights, S. Al Nussirat, et. al., "Across the soft gamma-ray regime: utilizing simultaneous detections in the Compton Spectrometer and Imager (COSI) and the Background and Transient Observer (BTO) to understand astrophysical transients," in *Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray*, Aug. 2024.  DOI: 10.1117/12.3020606.

Non-Refereed

R. Falco, A. Bulgarelli, N. Parmiggiani, et. al. (E. Neights 18th out of 29 authors), "COSIflow: a modular framework for orchestrating and monitoring automated scientific pipelines for the COSI mission," in *SPIE Software and Cyberinfrastructure for Astronomy IX*, 2026.  DOI: 10.1117/12.3105168.

- N. Parmiggiani, A. Bulgarelli, G. Panebianco, et. al. (**E. Neights** 5th out of 20 authors), “Localization and Confidence Region Estimation of Short GRBs with the COSI BGO Shield Using a HEALPix-Based Deep Learning Approach,” in *Proceedings of the ADASS XXXV conference*, Apr. 2026.  DOI: 10.48550/arXiv.2604.15119.
- E. Burns, J. Andrews, R. Szabo, et. al. (**E. Neights** 19th out of 26 authors), “The Heavy Element Enrichment History of the Universe from Neutron Star Mergers with Habitable Worlds Observatory,” *arXiv e-prints*, Jul. 2025.  DOI: 10.48550/arXiv.2507.09778.
- E. Burns, C. L. Fryer, I. Agullo, et. al. (**E. Neights** 63rd out of 85 authors), “Multidisciplinary Science in the Multimessenger Era,” *arXiv e-prints*, Feb. 2025.  DOI: 10.48550/arXiv.2502.03577.
- H. Yoneda, T. Siegert, I. Martinez-Castellanos, et. al. (**E. Neights** 12th out of 26 authors), “Imaging MeV Gamma-ray Lines with Advanced Image Reconstruction Framework for COSI,” in *39th International Cosmic Ray Conference*, Dec. 2025.  DOI: 10.22323/1.501.0891.
- J. Tomsick, S. Boggs, A. Zoglauer, et. al. (**E. Neights** 24th out of 74 authors), “The Compton Spectrometer and Imager,” in *38th International Cosmic Ray Conference*, Sep. 2024.  DOI: 10.22323/1.444.0745.
- I. Martinez-Castellanos, S. Gallego, C.-Y. Huang, et. al. (**E. Neights** 9th out of 75 authors), “The cosipy library: COSI’s high-level analysis software,” in *38th International Cosmic Ray Conference*, Aug. 2023.  DOI: 10.48550/arXiv.2308.11436.
- T. Akindele, T. Anderson, E. Anderssen, et. al. (**E. Neights** 91st out of 148 authors), “A Call to Arms Control: Synergies between Nonproliferation Applications of Neutrino Detectors and Large-Scale Fundamental Neutrino Physics Experiments,” in *Snowmass 2021*, Feb. 2022.  DOI: 10.48550/arXiv.2203.00042.
- T. Grégoire, H. Ayala Solares, S. Coutu, et. al. (**E. Neights** 11th out of 13 authors), “Model independent search for transient multimessenger events with AMON using outlier detection methods,” in *37th International Cosmic Ray Conference*, Mar. 2022, p. 934.  DOI: 10.22323/1.395.0934.

Presentations

Invited Talks & Seminars

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| Sep 2026 | Probing Gamma-Ray Bursts with Ultra-long GRB Observations and the Upcoming COSI Mission Transient Tea at Harvard & Smithsonian Center for Astrophysics (<i>Cambridge, MA</i>)
Gamma-Ray Burst from a Black Hole Falling into a Star Group Meeting at Massachusetts Institute of Technology (<i>Cambridge, MA</i>) |
| Aug 2026 | Probing Gamma-Ray Bursts with Ultra-long GRB Observations and the Upcoming COSI Mission Seminar at Cornell University (<i>Ithaca, NY</i>)
Probing Gamma-Ray Bursts with Ultra-long GRB Observations and the Upcoming COSI Mission Gravity, Astroparticle and Particle Physics Seminar at Penn State (<i>University Park, PA</i>) |
| Jun 2026 | COSI Observations of GRBs “Exploring the MeV Gamma-ray Sky with COSI” Special Session at the American Astronomical Society Meeting (<i>Pasadena, CA</i>) |
| Jun 2026 | Probing Gamma-Ray Bursts with Observations of Ultra-long GRBs and the Upcoming COSI Mission Seminar at California Institute of Technology (<i>Pasadena, CA</i>) |
| May 2026 | Studying GRBs with COSI Seminar at Flatiron Institute’s Center for Computational Astrophysics (<i>New York, NY</i>)
Gamma-Ray Burst from a Black Hole Falling into a Star Theoretical High Energy Astrophysics Seminar at Columbia University (<i>New York, NY</i>) |
| Mar 2026 | Discovery of a Gamma-Ray Burst from a Black Hole Falling into a Star American Physical Society Global Physics Summit (<i>Denver, CO</i>) |
| Oct 2025 | COSI GRB Group Update COSI Collaboration Meeting (<i>St. Louis, MO</i>) |
| Oct 2024 | COSI’s Spectral Analysis Software COSI Collaboration Meeting (<i>Berkeley, CA</i>) |

Presentations (continued)

- Oct 2024 **GRB Polarization with COSI** COSI Collaboration Meeting (*Berkeley, CA*)
- May 2024 **GRB Science Using COSI** Seminar at Oregon State University (*Corvallis, OR*)
- Apr 2024 **The COSI GRB Science Goals** “Exploring the MeV Gamma-ray Sky: The Past, Present, and Future” Special Session at the High Energy Astrophysics Division Meeting (*Horseshoe Bay, TX*)
- Jul 2024 **GRB Polarization with COSI** GRB Forum (*Athens, Greece*)

Contributed Talks

- Jun 2026 **Gamma-Ray Burst from a Black Hole Falling into a Star** American Astronomical Society Meeting (*Pasadena, CA*)
- Oct 2025 **GRB Localizations and Polarimetry with COSI** “Exploring the MeV Gamma-ray Sky with COSI” Special Session at the High Energy Astrophysics Division Meeting (*St. Louis, MO*)
- Jan 2025 **Gamma-Ray Burst Polarization with COSI** American Astronomical Society Meeting (*National Harbor, MD*)
- Aug 2024 **Studying GRBs Using COSI’s Polarization Measurements** AstroCon DC (*Washington, DC*)
- Jun 2024 **Studying GRBs Using COSI** Fermi Summer School (*Lewes, DE*)
- Apr 2024 **Studying Gamma-Ray Bursts Using COSI** High Energy Astrophysics Division Meeting (*Horseshoe Bay, TX*)
- Aug 2023 **Using COSI to Study GRBs** DMV Consortium Graduate Student Astrophysics Conference (*Washington, DC*)
- Aug 2022 **Scientific Pipeline Development for COSI** NASA Goddard Space Flight Center Astrophysics Summer Intern Symposium (*Virtual*)
- Sep 2020 **Archival Coincidence Analysis for IceCube Cascade Events & GRBs** IceCube Collaboration Meeting (*Virtual*)

Posters

- Oct 2025 **Gamma-Ray Observations of GRB 250702B: Gamma-Ray Burst from a Black Hole Falling into a Star** High Energy Astrophysics Division Meeting (*St. Louis, MO*)
- Mar 2023 **Introduction to COSI Data Analysis** High Energy Astrophysics Division Meeting (*Waikoloa, HI*)
- May 2020 **Photomultiplier Tube Pressure Testing for WATCHMAN** Pennsylvania State University Undergraduate Exhibition (*State College, PA*)
- Nov 2018 **The Study of Binary Neutron Star Mergers from Simulations** Pennsylvania State University Undergraduate Research Symposium (*State College, PA*)

Selected Press

- 2026 **BBC Sky at Night**
 skyatnightmagazine.com/news/grb-250702b-study-interview
- 2025 **National Geographic**
 nationalgeographic.com/science/article/gamma-ray-burst-black-hole-eats-star
- NASA**
 science.nasa.gov/science-research/black-hole-eats-star
- Sky and Telescope**
 skyandtelescope.org/astronomy-news/black-hole-eats-through-star-explodes-it-from-within
- GW Today**
 gwtoday.gwu.edu/black-hole-eats-star-student-charts-record-blast

Teaching

2022 **Teaching Assistant for Classical Mechanics**

Assisted with lesson delivery, graded, held weekly office hours, and led one independent lecture for an undergraduate electricity and magnetism class of 38 students.

2021 **Teaching Assistant for Electricity & Magnetism**

Assisted with lesson delivery, graded, and held weekly office hours for an undergraduate electricity and magnetism class of 25 students.

Outreach & Professional Development

2025 **Attendee of Summer School in Statistics for Astronomers**

2024 **Attendee of Fermi Summer School**

Volunteer for Astronomy Festival on the National Mall

2023-2025 **George Washington University Graduate Mentor** Supported junior graduate students in navigating the department, succeeding in coursework, and identifying research opportunities.

2023-2024 **Member of COSI Collaborations Connections Team** Addressed issues of diversity, equity, inclusion, and accessibility in COSI collaboration.

2020 **Volunteer for Penn State AstroFest**

2019-2021 **Safety Liaison for Penn State Social Dance Club**

Wrote safety and inclusivity policies, addressed concerns of members, and handled cases of misconduct.

2018 **Presenter at Penn State AstroFest**

Collaborations

COSI

Early Career Co-Lead for GRB Science Team, lead creation of transient observation simulations, make GRB science predictions, and develop spectral and polarization analysis software.

Fermi-GBM

Burst Advocate and led analysis of record-duration GRB 250702B.